

The Kripke–Wittgenstein Paradox Through the Lens of TNA: Rule-Following as Failure of Local Closure

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Abstract

The Kripke–Wittgenstein rule-following paradox argues that no finite set of behavioral facts can uniquely determine the rule being followed by a system. The same input-output history can be interpreted as infinitely many distinct functions. This paper proposes that the paradox can be reformulated as an instance of **Failure of Local Closure** under the Theory of Axiomatic Necessity (TNA). We argue that operational dynamics (N_0) are insufficient to determine the admissibility structure that legitimizes a rule. The distinction between addition and Kripke's "quus" function illustrates the necessity of a structurally external selector (N_1) that determines which interpretation is realizable. We demonstrate that the selector cannot be reduced to the operational domain without generating infinite regress, and we distinguish the TNA position from both eliminative materialism and functionalism. The paradox is thus revealed not as a linguistic puzzle but as a specific manifestation of a general structural principle: no operational system can fully derive the legitimacy of its own rule space from its internal dynamics alone.

1. Introduction

In *Wittgenstein on Rules and Private Language* (1982), Saul Kripke presents a skeptical challenge inspired by Ludwig Wittgenstein's later philosophy [1].

Suppose a person has only ever performed additions involving numbers smaller than 57. When asked:

$5 + 3 = ?$

the answer is:

However, consider the alternative function "quus" (denoted by the operator $\oplus$):

$$x \oplus y = \begin{cases} x + y & \text{si } x + y < 57 \\ 5 & \text{si } x + y \geq 57 \end{cases}$$

All previous calculations remain compatible with both addition and quaddition. The observable history does not determine which rule is being followed. The central question becomes: **What fixes the rule?**

Kripke's skeptical conclusion is that no finite collection of physical facts uniquely determines the intended meaning of a rule. The community's agreement, on Kripke's "skeptical solution," provides a pragmatic but not metaphysical answer: we follow addition rather than quaddition because that is what our community does.

This paper argues that the Kripke paradox is not merely a puzzle about language or meaning. It is a specific instance of a general structural principle formalized by the **Theory of Axiomatic Necessity (TNA)**: the **Failure of Local Closure** [2]. The paradox reveals that operational dynamics (N_0) cannot generate their own conditions of admissibility; a structurally external selector (N_1) is required to collapse the space of possible interpretations into a realized rule.

2. The Problem as Structural Underdetermination

Let:

$$H = \{(i_1, o_1), \dots, (i_n, o_n)\}$$

be the complete observable history of a computational system. There exists an infinite set of functions:

$$\mathcal{F} = \{f_1, f_2, f_3, \dots\}$$

such that:

$$f_k(i_j) = o_j$$

for all observed cases. Therefore:

$$H \not\Rightarrow f$$

The history underdetermines the rule. Kripke's conclusion highlights that no finite collection of physical facts uniquely determines the intended meaning of a rule. The same input-output history is compatible with infinitely many distinct functions, each diverging at some unobserved point.

3. The TNA Reformulation

TNA distinguishes between two structural domains [3]:

DOMAIN	DEFINITION	ROLE
N_0	Operational dynamics	Generates the space of possible trajectories
N_1	Admissibility structure	Selects which trajectories are realizable

Within this framework, circuit states, memory states, input-output histories, and computational transitions belong strictly to the operational domain (N_0). All are operational descriptions.

The choice between addition and quaddition is **not contained in the operational history itself**. The operational system generates multiple admissible interpretations:

$$\Sigma = \{f_{\text{add}}, f_{\text{quus}}, f_3, \dots\}$$

The paradox appears because N_0 alone cannot collapse Σ into a unique realizable rule.

4. Rule Selection as Failure of Local Closure

The paradox can be expressed as:

$$\dim(\Sigma) > 1$$

Multiple trajectories remain admissible. The system lacks a mechanism capable of selecting one interpretation from within its own operational domain. This is precisely the structure identified by TNA as **Failure of Local Closure** [2]:

■ *A local system cannot fully generate the conditions that legitimize its own interpretation.*

The question "*Is the machine adding or quadding?*" cannot be answered by inspecting only the machine.

5. The Necessity of N_1

TNA introduces a selector:

$$N_1 : \Sigma \rightarrow \sigma^*$$

which determines which trajectory becomes realizable. In the case of arithmetic:

$$N_1(\Sigma) = f_{\text{add}}$$

selects addition rather than quaddition. Importantly, N_1 is **not** another computational step inside the machine. If it were, the same paradox would simply reappear at the higher level. One would then need a second selector $N_1^{(2)}$ to interpret the first, generating an infinite regress.

5.1 The Collapse of Interpretation (Ψ)

TNA formalizes the transition from multiplicity to singularity through the **structural collapse operator** Ψ [3]:

$$\Psi : \Sigma \rightarrow \sigma^*$$

where:

$$\Psi(\Sigma) = N_1(\Sigma) = \sigma^*$$

The collapse is not a computational operation within N_0 . It is a **selection** that operates structurally external to the operational domain. Without Ψ , Σ remains a space of unrealized possibilities (with cardinality $\aleph[3]$); with Ψ , one trajectory becomes the realized interpretation σ^* .

6. The Regress Argument

Suppose, for the sake of contradiction, that the selection function $N_1 : \Sigma \rightarrow \sigma^*$ were computable within N_0 . Then there would exist an effective procedure $\Pi_{N_1} \subseteq N_0$ that selects the boundary condition (interpretation) for the system.

But the selection of an interpretation restricts the solution space of the system. Therefore, Π_{N_1} must contain a description of the conditions of possibility for its own operation:

$$\mathcal{D}(\Pi_{N_1}) \supseteq \{\Sigma_B\}$$

This yields a self-referential loop:

$$\Pi_{N_1} \rightarrow B \rightarrow \Sigma_B \rightarrow \mathcal{D}(\Pi_{N_1}) \rightarrow \Pi_{N_1}$$

Any procedure $\Pi_{N_1} \subseteq N_0$ that selects boundary conditions for S must itself be subject to boundary conditions for its own operation. Let B_Π be the boundary condition under which Π_{N_1} operates. Then B_Π must itself be selected by some $\Pi_{N_1}^{(2)}$, which requires its own $B_\Pi^{(2)}$, selected by $\Pi_{N_1}^{(3)}$, and so on:

$$\Pi_{N_1}^{(n)} \rightarrow B_\Pi^{(n-1)} \rightarrow \Pi_{N_1}^{(n-1)} \rightarrow \dots \rightarrow \Pi_{N_1}^{(1)} \rightarrow B \rightarrow \Sigma_B$$

This generates an infinite regress of selectors with no base case:

$$\Pi_{N_1}^{(1)} \leftarrow \Pi_{N_1}^{(2)} \leftarrow \Pi_{N_1}^{(3)} \leftarrow \dots \leftarrow \Pi_{N_1}^{(n)} \leftarrow \dots$$

Therefore:

$$\nexists \Pi_{N_1} \subseteq N_0 : \Pi_{N_1} \equiv N_1$$

Hence:

$$N_1 \not\subseteq N_0$$

in the structural sense: the function of selecting admissibility conditions is not computable within the state space whose admissibility is being determined [4].

7. The Calculator Example

Consider a calculator. From a purely physical standpoint, we observe:

Input: $5 + 3 \implies$ Output: 8

Nothing in this isolated event determines whether the device implements addition, quaddition, or any other compatible function. The semantic interpretation is supplied externally. The machine performs operations; the rule is imposed by a selector.

In TNA terminology:

- **The calculator is N_0 .**
- **The interpretation of the rule is N_1 .**

7.1 The Learning Extension

A critic might object: *"But what of a system that learns the rule? A child who learns to add does not have a 'designer' who imposes addition."*

This objection conflates training data with admissibility structure. In machine learning terms [5]:

- The training dataset is N_0 .
- The model weights after training are N_0 .
- The generalization to novel inputs requires a condition of realizability that is not contained in the data.

A neural network trained on examples of addition might generalize correctly to $57 + 68$, or it might not. The training history underdetermines the generalization, just as the finite history of calculations underdetermines the rule. The "correct" generalization is selected by an external admissibility structure: the mathematical framework within which the training was embedded, the teacher's intention, the cultural practice of arithmetic. These are not additional data points; they are **conditions of possibility** for what the data mean.

8. Parallel with Consciousness

The same structure appears in cognition. Suppose a brain enters a physical state B . A materialist description specifies neural firings, synaptic weights, and electrochemical dynamics. These are operational facts (N_0).

Yet the question remains: *Does state B mean addition or quaddition?*

The physical description alone appears insufficient to determine the normative content. The problem is structurally identical to the calculator case. The operational substrate does not uniquely determine admissibility.

9. Three Positions on N_1

The Kripke paradox forces a choice between three positions on the nature of rule-following:

POSITION	CLAIM	TNA ASSESSMENT
Eliminative Materialism	There is no N_1 ; "rules" are misdescriptions of N_0	Fails: cannot account for normativity or error

POSITION	CLAIM	TNA ASSESSMENT
Functionalism	N_1 is a pattern within N_0	Fails: generates regress; pattern requires prior admissibility
TNA	N_1 is structurally external to N_0	Accounts for selection without regress or mysticism

9.1 Against Eliminative Materialism

The eliminativist claims that there is no "fact of the matter" about which rule is followed, only physical transitions. But this cannot account for the **normativity** of rule-following: the distinction between correct and incorrect application. If there is no rule, there is no error. Yet error is constitutive of rule-governed behavior [1].

9.2 Against Functionalism

The functionalist claims that N_1 is a higher-order pattern within N_0 . But a pattern is itself underdetermined by the physical substrate: the same physical transitions can instantiate infinitely many patterns. The selection of which pattern is realized requires a selector, and we are back to the regress of Section 6.

9.3 The TNA Position

TNA does not postulate a supernatural entity. It postulates a **structural function**: the function of selecting between multiplicities. This function is instantiated physically (neurons, community practices, mathematical frameworks), but it is not identifiable with any particular physical instance and cannot be reduced to the operational domain it conditions. N_1 is not a "thing" but a **relation of selection** that operates asymmetrically: N_1 conditions N_0 , not vice versa.

10. Relation to the Boundary-Condition Framework

The paradox resembles a boundary-value problem in engineering [6]. The differential equation alone does not determine a unique solution; boundary conditions are required.

Likewise:

- N_0 specifies operational dynamics.
- But operational dynamics alone do not determine meaning.

- Meaning requires admissibility constraints analogous to boundary conditions.

The distinction between addition and quaddition is therefore not unlike the distinction between a simply supported beam and a fixed beam. The governing equation remains identical. The realizable solution changes because the admissibility structure changes.

In both cases:

N_0 = dynamics

B = boundary conditions (admissibility constraints)

N_1 = selector of boundary conditions

$(B, N_0) \rightarrow$ realized trajectory σ^*

11. Conclusion

The Kripke–Wittgenstein paradox can be interpreted as an epistemic instance of **Failure of Local Closure**. The observable behavior of a system does not uniquely determine the rule it follows because operational dynamics alone cannot generate their own conditions of admissibility.

Within TNA:

- N_0 produces operational trajectories.
- N_1 selects realizable interpretations.

The distinction between addition and quaddition is therefore not merely a linguistic puzzle. It reveals a deeper structural principle: **No operational system can fully derive the legitimacy of its own rule space from its internal dynamics alone.**

In this sense, the Kripke paradox becomes a specific manifestation of the more general TNA theorem: **The selector of admissibility cannot be reduced to the operational domain whose admissibility it determines.**

The skeptical challenge is not answered by finding more facts within N_0 , nor by retreating to conventionalism or community agreement (which are simply larger N_0 systems). It is answered by recognizing that the space of possible interpretations (Σ) requires a structurally external selector (N_1) to collapse into a realized rule (σ^*). This collapse is not magic; it is the fundamental operation by which possibility becomes actuality, multiplicity becomes singularity, and meaning becomes fixed.

12. Open Question: Experimental Testability

The TNA framework suggests an empirical research program [7]: How can we test whether a system has collapsed Σ into σ^* rather than into some other compatible interpretation?

In machine learning, this corresponds to the problem of **adversarial robustness** and **out-of-distribution generalization**: a model that has learned "addition" will generalize differently than one that has learned "quaddition," even if their training histories are identical up to some threshold. The test lies not in the training data (N_0) but in the **structure of generalization** — the signature of N_1 's selection.

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Appendix: Glossary of TNA Terms

TERM	DEFINITION
N_0	Operational domain: internal dynamics, equations, parameters, evolution rules
N_1	Selector domain: instance that imposes admissibility conditions, structurally external to the operational process
Σ	Space of possible trajectories/interpretations generated by N_0
σ^*	Realized trajectory/interpretation, selected by N_1
Ψ	Structural collapse operator: transition from multiplicity (Σ) to singularity (σ^*)
$\aleph$	Cardinality of the space of unrealized possibilities
Failure of Local Closure	Theorem: a system cannot generate the conditions of its own admissibility from within its operational domain